

Contents

Abstract	1
1. Introduction	2
2. Literature Review	3
2.1 Intro to Legume Seed Evaluation	3
2.2 Hyperspectral Imaging.....	3
2.2.1 Principles.....	3
2.2.2 Pre-Processing and Data Analysis Techniques	4
2.2.3 Applications	6
2.3 Hyperspectral Imaging in the food industry	6
3. Method.....	7
3.1 Preparing Seeds.....	7
3.2 Imaging the seeds.....	7
3.3 Germination Testing	8
3.4 Image processing.....	9
3.4.1 Converting from Volts to Reflectance.....	9
3.4.2 Masking Images	10
3.4.3 Creating Structure.....	13
3.4.4 PCA.....	14
3.4.5 K-Means Clustering.....	14
3.4.6 PLSDA.....	15
4. Results & Analysis	16
4.1 Germination Test	16
4.2 PCA	16
4.2.1 VNIR	17
4.2.2 SWIR.....	18

4.3 K Means Clustering.....	19
4.4 PLSDA.....	22
4.4.1 Germinated VS Non-Germinated	22
4.4.2 New VS Aged.....	26
5. Conclusion	30
References.....	32
Appendix	36
Appendix 1: Declaration of Authorship	36
Appendix 2: Radiance to Reflectance function	36
Appendix 3: Unfold Function.....	41
Appendix 4: PCA Function	41
Appendix 5: SNV Function	49
Appendix 6: MSC Function	49
Appendix 7: K-means Function	50
Appendix 8: PLSDA with Cross Validation Function	61
Appendix 9: PLSDA Function	61
Appendix 10: Cumulative Variance Graphs.....	65

Table of Figures

Figure 1: Setup of apparatus for imaging seeds	8
Figure 2: RGB image of kidney bean in the VNIR region (left) and SWIR region (right)	10
Figure 3: Spectra for different pixels in a VNIR image (left) and SWIR image (right)	11
Figure 4: Ratio image for kidney bean in the VNIR region (left) and SWIR region (right)	11
Figure 5: First mask for VNIR (left) and SWIR (right) images.	12
Figure 6: Example PCA score images for making the second and third mask for the seeds. ..	12
Figure 7: Example histogram (left) for PCA score image to make second mask (middle) and third mask (right).	13
Figure 8: Example of final mask for VNIR (left) and SWIR (right) images.	13
Figure 9: Ground truth images for (a) mixed bean sets with new and old seeds (green = new, red = old), (b) mixed set 1 germinated (green) and non-germinated (red), (c) mixed set 2 germinated (green) and non-germinated (red)	17
Figure 10: PCA score images for VNIR images of mixed seeds set 1 for different pretreatments.	18
Figure 11: PCA score images of SWIR images of mixed seeds set 1 for different pretreatments.	19
Figure 12: K-means clustering resulting images, showing the different clusters with different colors.....	21
Figure 13: Germination: Mean Class Accuracy (left) and cumulative variance (right) for PLSDA with cross validation (VNIR)	22
Figure 14: Germination: Mean Class Accuracy (left) and cumulative variance (right) for PLSDA with cross validation (SWIR)	23

Figure 15: Germination: Mean Class Accuracy (left) and cumulative variance (right) for PLSDA with cross validation (VNIR+SWIR combined)	23
Figure 16: Germination: Correct classification from PLSDA for each latent variable with different pretreatments (VNIR)	24
Figure 17: Germination: Correct classification from PLSDA for each latent variable with different pretreatments (SWIR)	25
Figure 18: Germination: Correct classification from PLSDA for each latent variable with different pretreatments (VNIR+SWIR combined)	25
Figure 19: AGE: Mean Class Accuracy PLSDA with cross validation (VNIR)	26
Figure 20: AGE: Mean Class Accuracy for PLSDA with cross validation (SWIR)	27
Figure 21: AGE: Mean Class Accuracy for PLSDA with cross validation (VNIR+SWIR Combined)	28
Figure 22: AGE: Correct classification from PLSDA for each latent variable with different pretreatments (VNIR)	28
Figure 23: AGE: Correct classification from PLSDA for each latent variable with different pretreatments (SWIR)	29
Figure 24: AGE: Correct classification from PLSDA for each latent variable with different pretreatments (VNIR+SWIR Combined)	30

List of Tables

Table 4.1: Germination Test Results	16
---	----

Abstract

In this study, hyperspectral imaging (HIS) was used to classify kidney bean seeds based on their age and viability. A total of 400 kidney bean seeds were used, which were a mix of 201 new seeds and 199 aged seeds which were aged for 2 years naturally. HySpex hyperspectral cameras in the visible-near infrared (VNIR)(406nm-998nm) and short-wave infrared (SWIR)(951nm-2518nm) regions were used to image sets of 25 seeds. Two hundred seeds (101 new, 99 old) were used in a germination test to find the germination percentage of both old and new seeds, and to see if the hyperspectral images could differentiate viable and non-viable seeds. PCA, K-means clustering, PLDA with “Leave One Out” cross validation and normal PLSDA were performed on the mean spectra of all beans. Using the PC score images from PCA with SNV or MSC, the new and old beans could be differentiated in the VNIR region. K means was able to correctly cluster new and old beans in the VNIR region with an accuracy of 89.9% when pretreated with SNV or MSC. Using PLSDA, a model with 86.2% correct classification accuracy was achieved in the SWIR region for viable and non-viable seeds with no pretreatment. Normal PLSDA achieved a model with 100% correct classification between new and old seeds in both the VNIR and SWIR region using Savitzky-Golay 1st derivative with SNV as a pretreatment.

1. Introduction

Legume seeds play a major role in the world's agriculture, due to their high nutritional value, use for chronic disease prevention [\[1\]](#), and use for soil fertility [\[2\]](#). Using the best quality seeds is of the utmost importance to insure a good crop yield. Therefore, tests must be carried out to investigate the legume seeds germinability, viability and vigour. In traditional seed evaluation methods, such as germination testing, or testing for moisture content, the seed is often destroyed and cannot be used again. Recently hyperspectral imaging has emerged as promising technology where seeds can be evaluated without damage as it is a non-destructive technique [\[3\]](#). It can give us information about the chemical composition, moisture content and structural properties of seeds using the spectra obtained. In this investigation kidney beans are imaged using Hypspx cameras in the visible-near infrared (VNIR)(406nm-998nm) and short-wave infrared (SWIR)(951nm-2518nm) regions, and multiple processes are carried out to identify and classify differently aged seeds, and seeds with the potential to germinate or not germinate.

2. Literature Review

2.1 Intro to Legume Seed Evaluation

Germination is recognized as the emergence of the radical root from the seed at the minimum. The disadvantages of using some traditional methods of seed evaluation are that they require the germination of the seed or for the complete destruction of the seed. This is of course undesirable. A traditional test for seed evaluation is one such as a simple germination test. This is where a large sample of seeds, greater than 400 sample size, is given the best conditions for germination, and the number of seeds that germinate is the germination percentage [4].

Another seed evaluation metric is the moisture content of the seeds. This is expressed as a percentage of wet weight in a sample of seeds. There are multiple ways that seeds can be tested for moisture. In [5] used the microwave method and oven method to determine the moisture content of maize seeds. The oven-drying method proved to be the most accurate. This is where the seeds are placed in a container and dried in an oven at specified time and constant temperature. The container is measure first (W1), then with the seeds in it (W2), and finally the container with the seeds after they have been dried (W3). The formula to determine the seeds moisture content is shown below in eq.1.

$$\text{moisture content percentage} = \frac{W2-W3}{W2-W1} * 100 \quad (\text{Eq.1})$$

2.2 Hyperspectral Imaging

2.2.1 Principles

A normal multispectral camera typically takes a photo using a small number of wavebands in the electromagnetic spectrum, usually capturing the three main primary colours (red, green,

blue) and some of the infrared region [6]. These bands are then combined to form multispectral image. For hyperspectral imaging, the image is captured in hundreds of much narrower wavebands (around 5-20nm wide) [7]. This makes hyperspectral imaging more sensitive to slight changes in the reflected/absorbed energy from the sample. Line scan hyperspectral imaging is a major technique for hyperspectral imaging [8] [9]. The sample is placed on a moving stage and is imaged line by line. By doing this, a data cube can be formed. It contains spatial data in the x-y plane, and wavelength data in the z direction, i.e., there a spectrum of wavelengths for each pixel. Using this data, the image can be analysed at different known wavelengths.

2.2.2 Pre-Processing and Data Analysis Techniques

A raw hyperspectral image is often useless without some form of pre-processing. In many hyperspectral images there will be stochastic noise, dead pixels, and unwanted background to name but a few [10]. To remove these unwanted artifacts, preprocessing methods such as masking is used. A mask is a binary image of the hyperspectral image, with the unwanted pixels set to 0, and the rest set to 1. When the hyperspectral image is multiplied by the mask, it creates a new image with only the useful pixels. Before more processing can be take place, the hyperspectral image must be unfolded. Unfolding is taking a $n * m * \lambda$ three dimensional HSI and creating a $(n * m) * \lambda$ two-dimensional dataset, where each row represents the spectrum of its corresponding pixel in the HSI. These spectra in the unfolded image are then ready to be processed.

PCA (principal component analysis) is a multivariate data analysis technique that combines existing variables to create principal components (PC's) and is great for data size reduction [11]. These principal components summarize the data and are created to maximise variance.

The first principal component has the most variance, and the variance diminishes quickly after this. The PCA transformation can be summarized by equation 2 below [\[12\]](#).

$$X = (S * L) + E \quad (\text{Eq.2})$$

, where X is the unfolded data set, S is the scores for each PC, L is the loading for each PC and E is the eigenvalues for each PC. Scores are projections of the original data and is the unfolded data is of size $j * \lambda$, the score for each PC is $j * A$, where A is the number of significant PC's. The loadings are weights assigned to X to produce the PCS's and are of size $A * \lambda$, and eigenvalues represent the variance of each PC. Using this information, we can bring over 200 wavelength variables to around 5 PC's.

Another technique that can be used to cluster related spectra is k-means clustering. K-means clustering is an unsupervised partitional method of clustering. It creates a set number of cluster centres 'k' and finds the distance from each data point (spectrum) to the centre of each cluster. Spectra are then assigned to their closest cluster, and new cluster centres are found depending on the mean of each previous cluster's centre. This process is repeated until convergence [\[13\]](#).

Partial least squares discriminate analysis (PLSDA) is another technique that can be used to classify data in many areas such as food analysis [\[14\]\[16\]](#). It is frequently used to classify spectra in a hyperspectral image as it is great for dimension reduction and does not require the data to be in any specific distribution [\[15\]](#). In [\[16\]](#), PLSDA was used to create a classification model to predict viability and vigour in muskmelon seeds from NIR hyperspectral images, achieving 94.6% classification accuracy.

2.2.3 Applications

There are many applications for hyperspectral imaging in today's world. For example, it is commonly used for organizing and sorting waste [\[17\]](#). Normally sorting waste would be a tedious and time-consuming job, but by using hyperspectral imaging, different materials can be identified quickly and sorted, saving time and money. Hyperspectral imaging is also used in the food industry [\[17\]](#) [\[18\]](#). It can be used to detect contaminants in foods, dehydration, ripeness, and monitor food processing [\[18\]](#). As it is a non-destructive method, it greatly reduces food waste.

2.3 Hyperspectral Imaging in the food industry

As mentioned above, hyperspectral imaging is becoming a commonly used tool for analysis in the food industry. Many studies have been undertaken using hyperspectral imaging on different types of seeds for different purposes. In [\[19\]](#), the effect that storage had on germination percentage and energy, and simple vigour index was studied using hyperspectral imaging. During the investigation the seeds were artificially aged, using one of the most common methods of storing them in a high temperature (around 40°C) and high humidity (100%) for up to 72hrs [\[19\]](#) [\[20\]](#) [\[21\]](#). In [\[22\]](#), soybeans were investigated using hyperspectral imaging to detect lipid changes during natural aging as well as accelerated aging and found that the lipid changes during the two different aging techniques was not the same. Hyperspectral imaging was also used in [\[23\]](#) to classify maize seeds of different age. The classification models were built using least squares support vector machine (LSSVM). A classification accuracy of 94.4% was achieved using these models. In [\[24\]](#), HIS was used to predict the cooking times of different beans that were naturally aged under different types of conditions, and the NIR spectra of the beans proved to be the most useful.

3. Method

3.1 Preparing Seeds

Kidney beans and aduki beans were the chosen beans to be investigated. The new seeds were obtained from a local organic food store, while the older seeds were purchased from the same store 2 years previously. It was ensured that they were the same kind of bean and same brand. The beans were then sorted into sets of 25 seeds, as this was the maximum that could be imaged at once using the Hyspex hyperspectral imaging equipment. Gloves were worn at all times while handling seeds to prevent contamination. Any visually bad looking seeds were discarded, and best efforts were made to keep the size and shape of both old and new seeds uniform. Individual plastic bags were then labelled for each bean. The labels followed a specific naming template so that each bean could be monitored, which was 'type of bean_set number _bean number'. So, for example, 'newbean_set1_4' would be the fourth bean in set 1 of the new beans.

3.2 Imaging the seeds

The beans were laid on a white tile on the Hyspex moving platform, the tile serving the purpose of the image background. They were arranged in a 5x5 grid pattern, and labelled 1-25 from left to right, top to bottom. A 100% reflectance reference tile was placed to the left of the sample so that it was at the start of every image. The final setup is shown in fig.1.

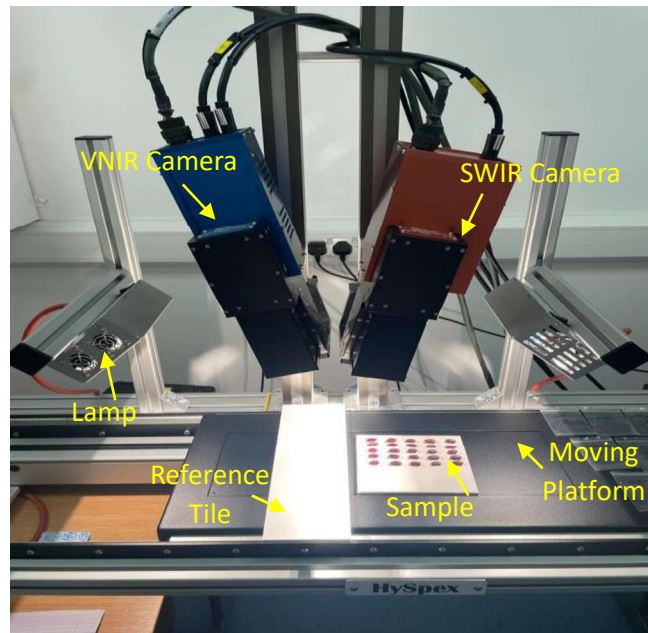


Figure 1: Setup of apparatus for imaging seeds

The first images taken were of three sets of new seeds on their own, and then three sets of old seeds on their own. Two sets of mixed beans (old and new) were also imaged, with beans 1-13 being new seeds and 14-25 being old seeds. These initial 200 kidney beans (101 new, 99 old) were then used in a germination test. A further 100 new and 100 old kidney beans were then imaged to increase the data set to work with. These extra 200 seeds were not tested for germination.

For the aduki beans, 200 beans had been initially imaged (100 old and 100 new), but once these were used for germination testing the majority was destroyed with mould and the samples were discarded. The kidney beans then became the only bean to be investigated.

3.3 Germination Testing

Once a seed from the initial 200 beans were imaged, they were then transferred into their individual sealed plastic bags. Each seed was then wrapped in a water wipe and placed back into its individual bag and sealed. The water wipe is what provided a growing medium and a source of water, while sealing the bag prevented any loss of moisture. The sealed bags

were then placed into an *Early Grow* heated medium propagator, to keep a constant environment for the seeds at 16-21°C. Each seed was then inspected after two days to check for signs of germination, and again after 7 days. The number of seeds that germinated in each set was noted and also the number of each seed for each group, to link the germination tests with the hyperspectral images. After the 7-day period for each seed set, the set was discarded, and a new set was imaged and prepared for germination.

3.4 Image processing

3.4.1 Converting from Volts to Reflectance

After the hyperspectral images were taken on the HySpex software, they were transferred to an external storage device and then transferred onto a laptop. HySpex captures two images for each sample, a visible-near infrared (VNIR) image and a short-wave infrared (SWIR) image. The images are originally in *.hyspex* file format and measured in volts. They first had to be transferred to *.img* file format (measured in radiance) and then converted to reflectance. To convert from *.hyspex* to *.img* (volts to radiance) the native hySpex software, *HySpexRadV2.0*, was used for each image. The resulting *.img* files were saved in the desired folder along with the *.hdr* files for every *.img* file that was generated while making the radiance images. To transform the radiance images from radiance to reflectance, MATLAB was used. The function *hsf_rad2reflectance* (APPENDIX 2) was the main function used for this purpose. Its inputs were the *.img* file name and the white reference MATLAB file name. This function was used in a loop to open an RGB image of each input *.img* file 25 times, and a new bean was cropped each time from 1 to 25. These cropped images were then saved each loop iteration using the same naming template as the plastic bags the seeds were kept in. Each bean was cropped individually for two reasons. To have

independent images of each bean, and because the whole image of the 25 beans for each set was too large to work with on its own.

3.4.2 Masking Images

Once all 500 seeds had been cropped and transformed into reflectance images, there was two types of images for each bean, SWIR and VNIR. An example of an RGB image of one of these beans is shown below in fig.2.



Figure 2: RGB image of kidney bean in the VNIR region (left) and SWIR region (right)

For both types of images, the background was first removed using the first mask. This was created using a ratio image. This ratio image was created by dividing the images by themselves at different wavelengths. In fig.3, the SWIR and VNIR spectrum for the different beans in the first set of new beans, the shadow of the beans and the background tile are shown. The resulting ratio image for both is shown in fig.4.

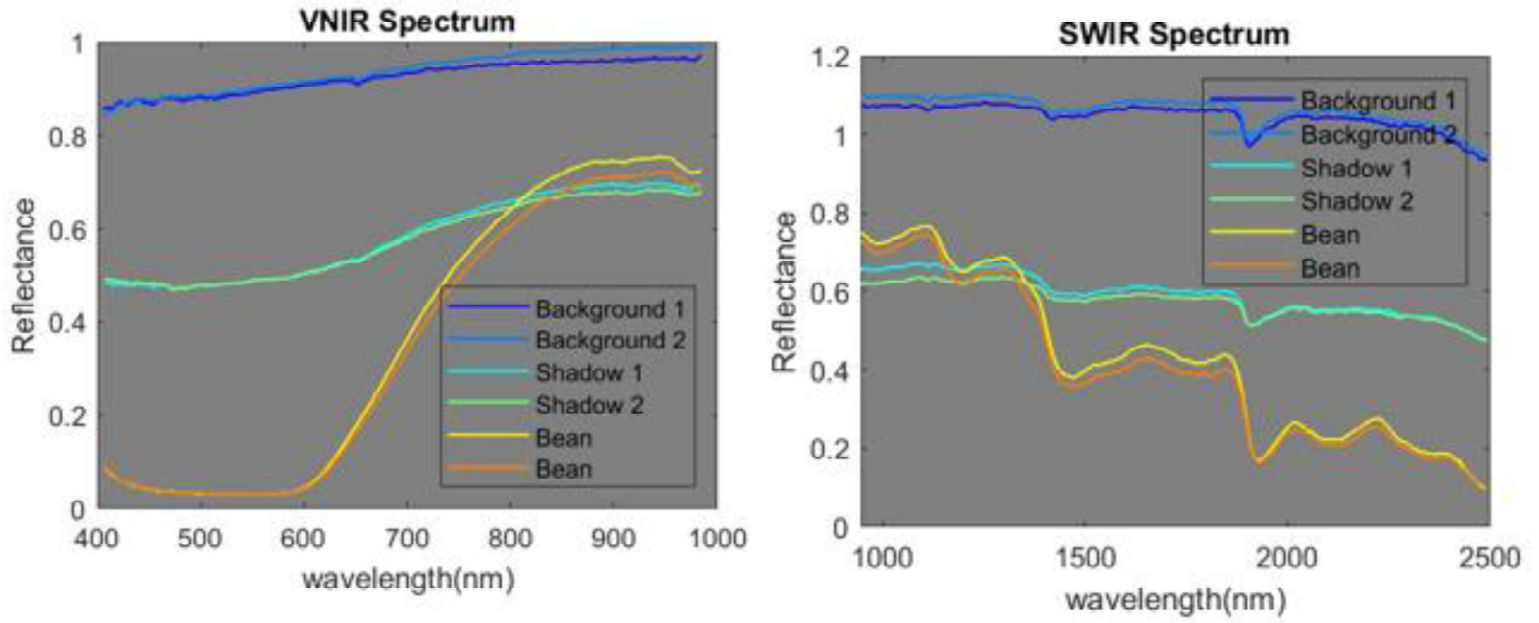


Figure 3: Spectra for different pixels in a VNIR image (left) and SWIR image (right)

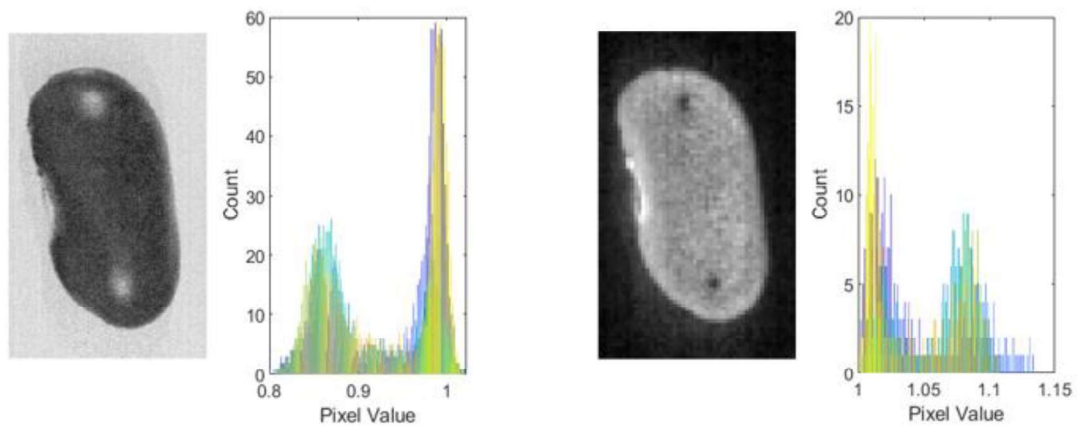


Figure 4: Ratio image for kidney bean in the VNIR region (left) and SWIR region (right)

From these images thresholds were set on the image's histogram and anything on one side of the threshold was either 0 or 1, depending on the mask needed. The images were then multiplied by their respective masks. An example of a first mask for a single bean in VNIR and SWIR images is shown in fig.5.

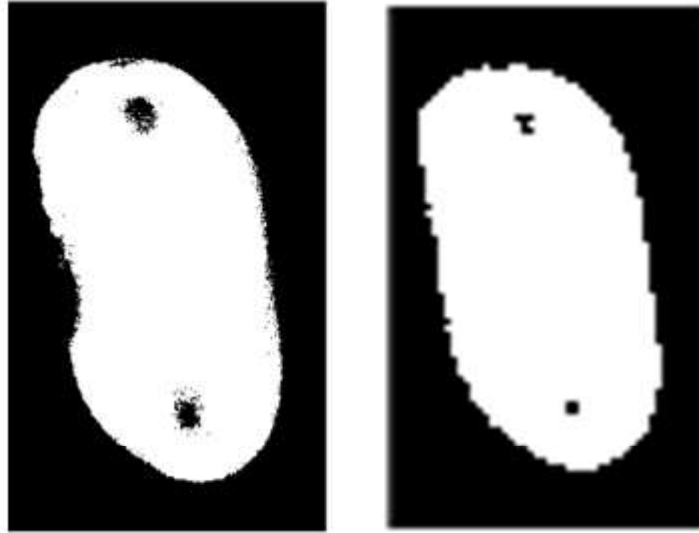


Figure 5: First mask for VNIR (left) and SWIR (right) images.

To obtain the remaining masks, PCA was applied to the first masked images and the histogram of the score images were used to create the masks. An example of the first five score images in of the VNIR image (fig.6) and a second and third mask for a cropped kidney bean are shown below in fig.7. Combining all these masks give the final overall mask for each cropped kidney bean, with an example shown in fig.8.

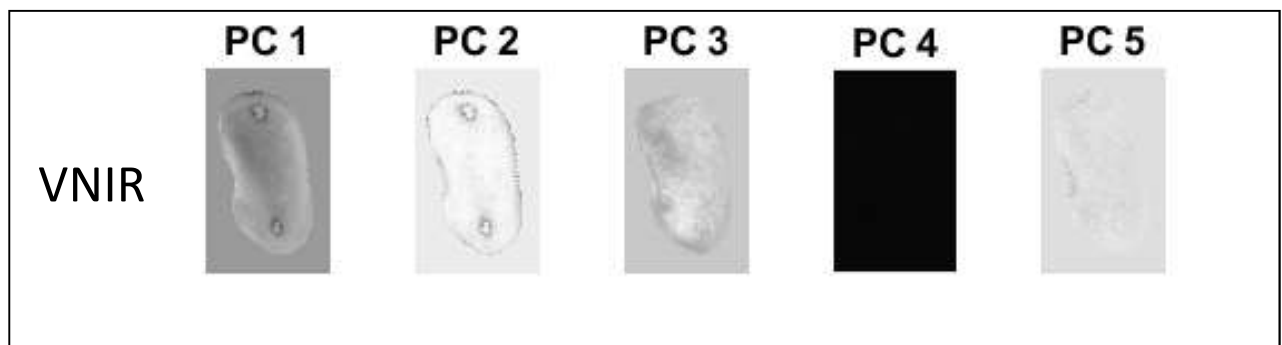


Figure 6: Example PCA score images for making the second and third mask for the seeds.

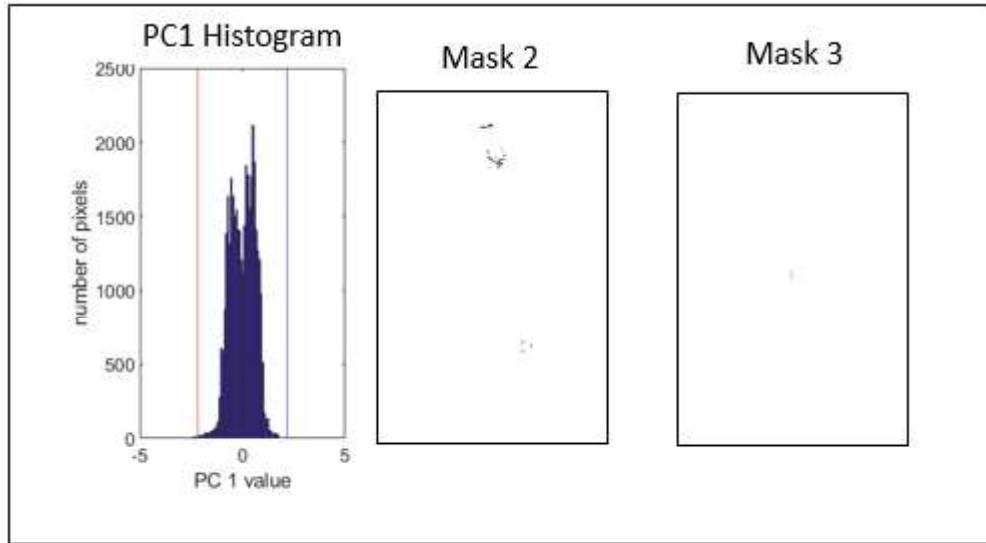


Figure 7: Example histogram (left) for PCA score image to make second mask (middle) and third mask (right).

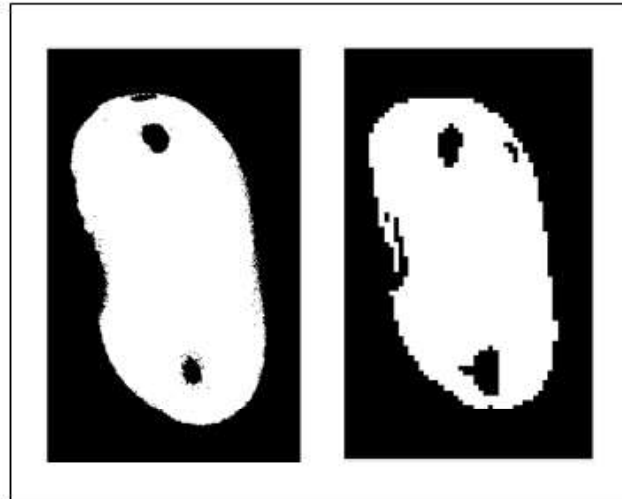


Figure 8: Example of final mask for VNIR (left) and SWIR (right) images.

3.4.3 Creating Structure

Once all 400 kidney beans had been separated, structure arrays were created to store the mean spectrum of each image in the SWIR and VNIR regions, as well as information such as whether each bean germinated or not, or whether it was a new or aged bean. The mean spectrum of each image was found by unfolding the image and finding the mean in the first dimension of the unfolded image. In the structure, the VNIR and SWIR wavelengths were

combined to create a full range of wavelengths. This was done by using the full VNIR range and concatenating it with the SWIR range from the ninth index onwards. Two structures were created overall. The first containing the initial 200 kidney beans (101 new and 99 old) that were tested for germination. The second structure array contained the remaining 200 kidney beans (100 old and 100 new).

3.4.4 PCA

PCA was then applied to the images of the mixed seed sets that were used in the germination test. Once the images were masked as above, the images were then unfolded using the *unfold* function in appendix 3. The function *PCA* (Appendix 4) was used on this masked unfolded dataset in both the SWIR and VNIR regions. This function returned the PCA scores, loadings and eigenvalues needed for analysis. The variance for each PC was then plotted, as well as the loadings. The score images for each PC were then reconstructed to see if any differentiation was visible between new and old, or germinated and non-germinated seeds. PCA was then performed on the masked and SNV data and masked and MSC data. To perform SNV and MSC, the *SNV* and *MSC* functions (Appendix 5-6) were used on the masked unfolded spectra.

3.4.5 K-Means Clustering

K-means clustering was then performed on the mixed seed sets using the function *k_means* (Appendix 7) was used. The number of classes was set to two: young and old, or germinated and non-germinated. This function was used on the masked spectra, masked and SNV, and masked and MSC spectra in both SWIR and VNIR. The clustered data was then reconstructed into an image to show the clustering results on the seeds.

3.4.6 PLSDA

Finally, PLSDA was performed on the unfolded spectra. Two tests were performed using PLSDA, one to classify germinated and non-germinated, another to classify aged and new seeds. To begin, the kidney bean data structure with the 200 beans used in germination testing was used. The first 75 seeds of the new and old were taken from the structure (150 total) and were labelled as the calibration set for the PLSDA algorithm. The last 25 of the new and old seeds from the same structure were labelled as the test set. Vectors were then made with corresponding labels to each calibration and test set to say whether each sample germinated or not, and with each seeds designated number. The calibration set was first taken and used for PLSDA cross-validation with 'leave one out' (LOO) using the function *Npls_da_N_SEP_pre_randCV_SS_LOSO* (Appendix 8). This is where one individual spectrum in the calibration is removed to be used as a test and the PLSDA model is built using the rest of the calibration set and is tested using the test spectrum. This is repeated for every spectrum in the calibration set. In this function seven different pretreatments are used to compare their effect for up to 20 latent variables. These were no treatment, SNV, Savitzky-Golay (1st derivative), Savitzky-Golay (2nd derivative), Savitzky-Golay (1st derivative) with SNV, Savitzky-Golay (2nd derivative) with SNV, and MSC. The mean class accuracy and variance was then found for each pretreatment and latent variable.

The full calibration and test sets were then used for normal PLSDA, using the function *Npls_da_N_SEP_pre_SS* (Appendix 9). The same seven pretreatments were used again and up to 20 latent variables were used. The confusion matrices were then used to find the correct classification accuracy. This PLSDA process was repeated using the structure with 200 non germination tested seeds to classify seeds based on age.

4. Results & Analysis

4.1 Germination Test

Table 1 shows the results from the 200 seeds that were germination tested. It was calculated that out of 101 new seeds, 13 had not germinated, giving a germination percentage of 87.1%. From the 99 aged seeds, 40 did not germinate, giving a germination percentage of 59.6%. This shows a clear difference in the germination percentage of the new and aged seeds, and therefore the age of the seed can be correlated to its vigour and viability.

Table 4.1: Germination Test Results

<u>Set</u>	<u>Non-Germinated seeds</u>	<u>Set</u>	<u>Non-Germinated seeds</u>
Mixed Set 1	9,19,22,23,14,17,21	New Set 3	5,18,2,17
Mixed Set 2	12,23,24	Old Set 1	11,7,19,23,21,15,3,25,22,14,13,16,12
New Set 1	2,12,9,8,11,17,24	Old Set 2	5,10,16,19,18,7,8,3
New Set 2	3,14	Old Set 3	4,9,2,14,19,7,11,24,22,5,15

4.2 PCA

Principle component analysis (PCA) was applied to the two sets of mixed seeds to investigate whether it could be used to differentiate between old and new seeds, or even determine whether seeds would germinate or not. Below the VNIR and SWIR PCA results are discussed. Ground truth images for the mixed seed sets is shown in fig.9, with fig.9(a) showing the new and old seeds, and fig.9(b-c) showing germinated and non-germinated. For both mixed sets, seeds 1-13 were new and 14-25 were aged, so only one ground truth image is needed.

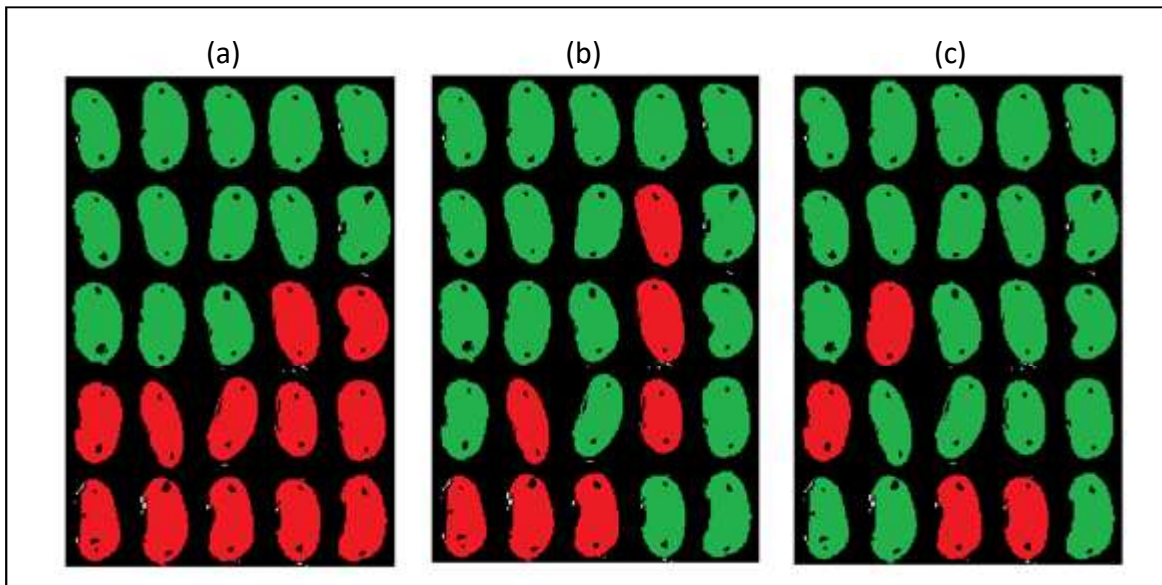


Figure 9: Ground truth images for (a) mixed bean sets with new and old seeds (green = new, red = old), (b) mixed set 1 germinated (green) and non-germinated (red), (c) mixed set 2 germinated (green) and non-germinated (red)

4.2.1 VNIR

In fig.10 below, the score images from performing PCA on the VNIR images of the mixed seed sets are shown. PCA was applied to the images with just the mask, the mask with a pretreatment of SNV, and the mask with a pretreatment of MSC. It is clear that using the raw masked image does not give any significant differentiation between new and old beans or germinated and non-germinated beans. However, using SNV and MSC give some differentiation between old and new beans, with the new beans be generally brighter in colour in the first score image for both. There is no clear separation of seeds beyond the PC1 images for both pretreatments.

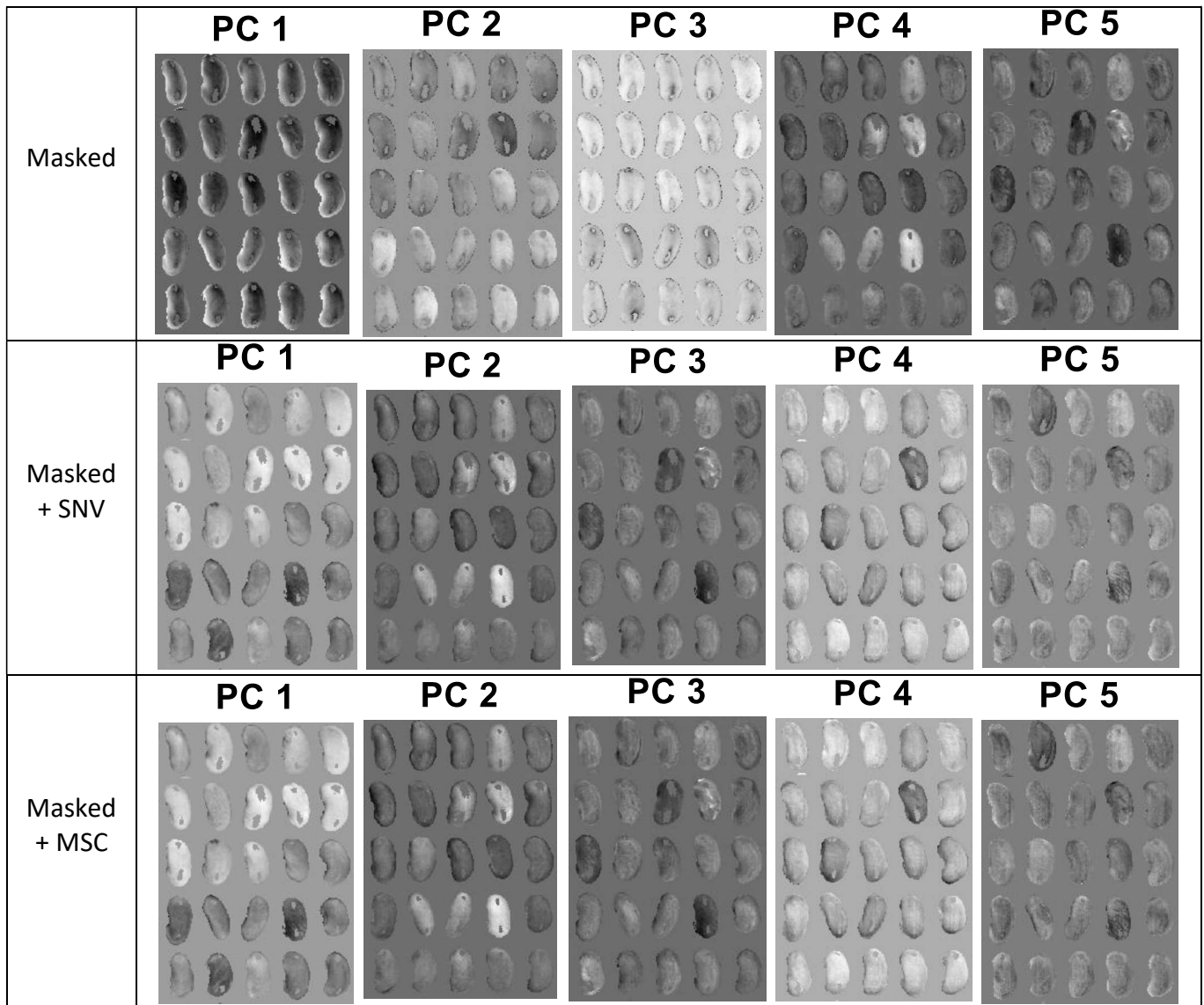


Figure 10: PCA score images for VNIR images of mixed seeds set 1 for different pretreatments.

4.2.2 SWIR

The PCA results for the SWIR images are shown in fig.11. As in the PCA score images for the VNIR images, PCA on masked only SWIR images did not produce any images that showed a differentiation between different bean classes. But unlike the VNIR results, using SNV and MSC did not make any significant separation of the classes in any of the score images shown.

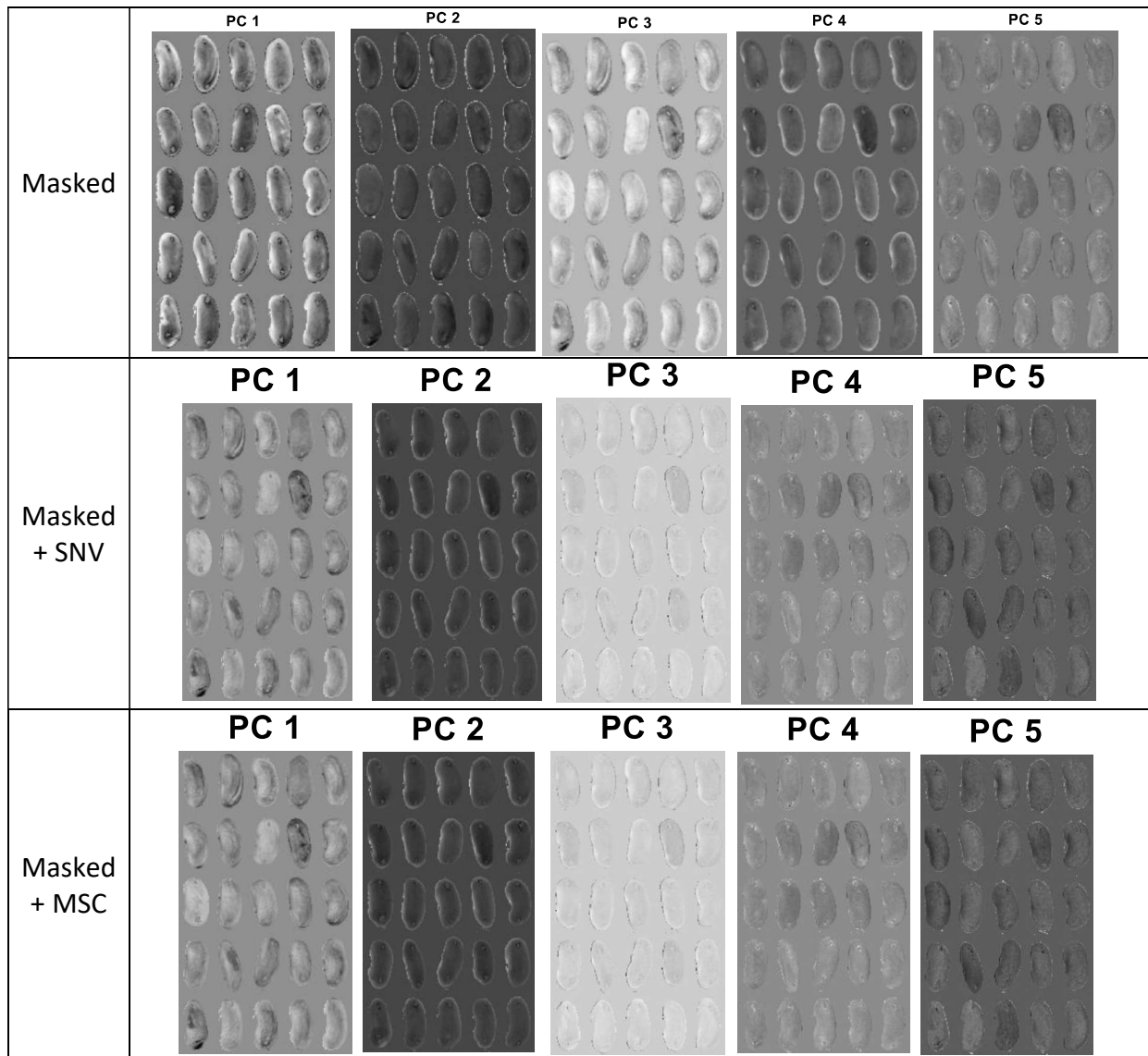


Figure 11: PCA score images of SWIR images of mixed seeds set 1 for different pretreatments.

4.3 K Means Clustering

K-means was also applied to the missed seed images obtained for the germination tests. The k-means clustering was performed with 2 clusters, for new and old seeds, or germinated and non-germinated. It was performed three times for each image (VNIR and SWIR) for both sets of mixed seeds; with just the mask, with mask and a pretreatment of SNV and with mask and a pretreatment of multiplicative scatter correction (MSC). The results are shown in fig.12. below.

Beginning with the VNIR images, for both sets of mixed seeds, performing k-means clustering on the masked spectra alone did not show any useful clusters. Only when SNV or MSC was used did k-means clustering work to separate the spectra. The SNV and MSC VNIR clustered images show most of the new and old seeds clustered correctly. For SNV, it clustered 12 of 13 new beans correctly and 11 of 12 old beans correctly for the first set of seeds, and 12 of 13 new beans correctly and 10 of 12 old beans correctly. So, for SNV this equates to 89.9% correctly clustered when compared to the ground truth image shown in fig.9(a). The results for the MSC clustered images were found to be the same as for SNV, with 89.9% seeds being correctly clustered. There is no clear clustering of the germinated and non-germinated seeds in the VNIR clustered images, only that of new and aged seeds.

The SWIR k-means clustered images are quite different than that of the VNIR. For the masked only images, also did not show any signs of helpful clustering although neither do the SNV and MSC images. Both methods (SNV & MSC) give similar results to each other, but do not correlate to the ground truth images shown in fig.9(a-c). This shows that using k-means clustering on the SWIR mixed seeds does not yield any helpful information.

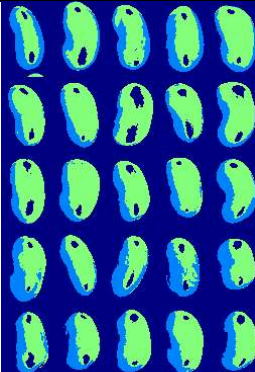
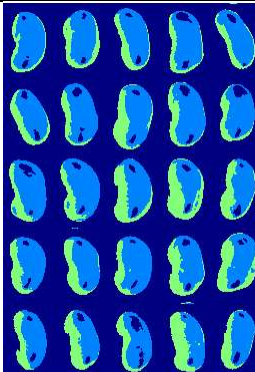
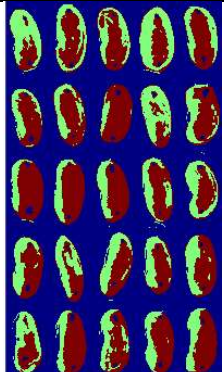
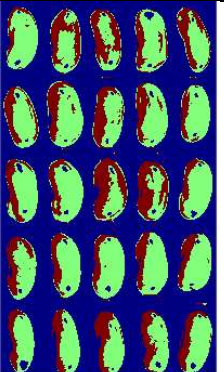
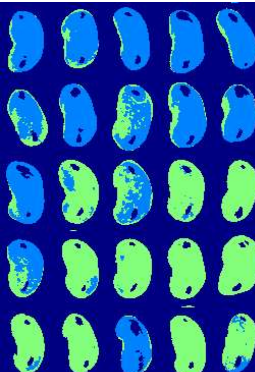
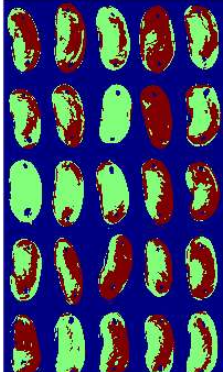
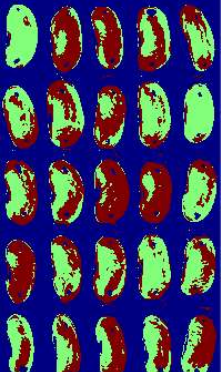
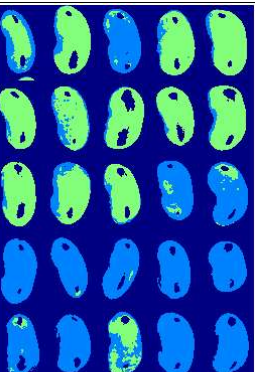
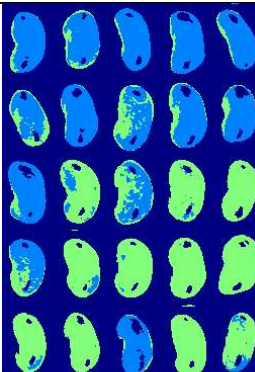
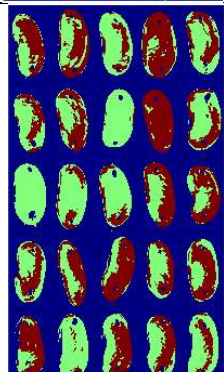
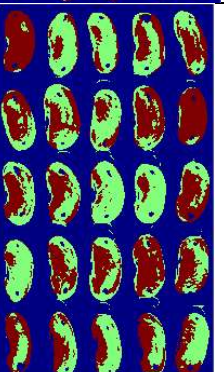
	VNIR (mixed set 1, set 2)		SWIR (mixed set 1, set 2)	
MASKED ONLY				
SNV				
MSC				

Figure 12: K-means clustering resulting images, showing the different clusters with different colors

4.4 PLSDA

PLSDA was used on the spectra of all the kidney beans. It was first used on the initial 200 seeds data from the structure containing the germination tested seeds, and then for the extra 200 seeds that were imaged.

4.4.1 Germinated VS Non-Germinated

4.4.1.1 PLSDA with Cross Validation

Fig.13 below shows how the mean class accuracy, from PLSDA with cross validation, changes with the number of latent variables. For the VNIR region, it can be seen that using a pretreatment of MSC gives the highest mean class accuracy of 89% at 20 latent variables. There is also a mean class accuracy of 72.9% at only 8 latent variables, which is compromise between accuracy and model complexity and computational cost. Using 8 latent variables for the VNIR region gives 81% percent in the variance using second derivative and SNV together as pretreatment.

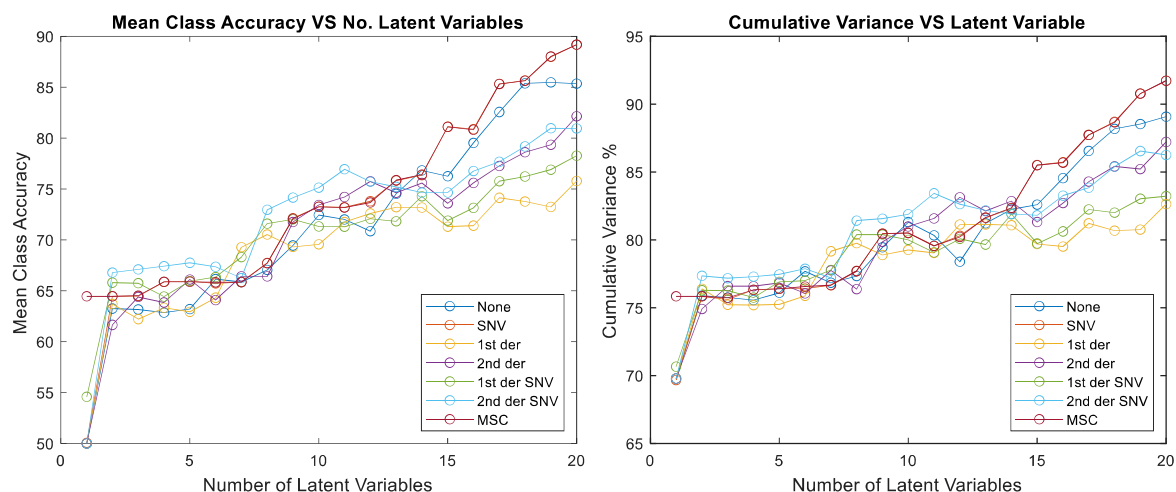


Figure 13: Germination: Mean Class Accuracy (left) and cumulative variance (right) for PLSDA with cross validation (VNIR)

In the SWIR region, using only 5 latent variables for the PLSDA with cross validation gives a mean class accuracy of 77% and this also covers 83% of the variance, as can be seen in fig.14 andx below. This is more accurate than using PLSDA on the VNIR region as before.

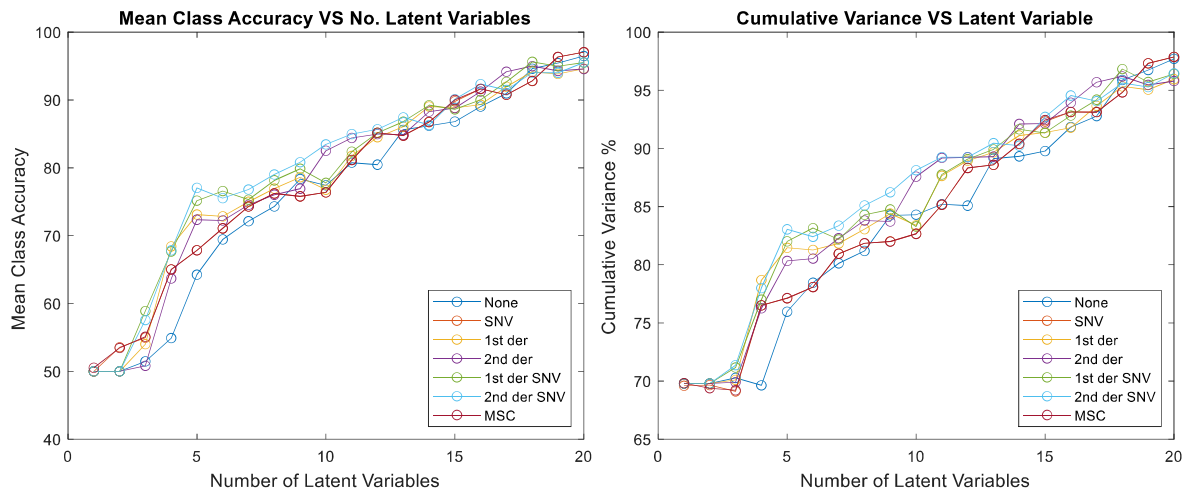


Figure 14: Germination: Mean Class Accuracy (left) and cumulative variance (right) for PLSDA with cross validation (SWIR)

The VNIR and SWIR regions were then combined and PLSDA was performed on the full wavelength range of the combined spectra. The results are shown in fig.15. If 9 latent variables are used for PLSDA with cross validation on the combined wavelength range, then 81% accuracy can be achieved having 87% variance.

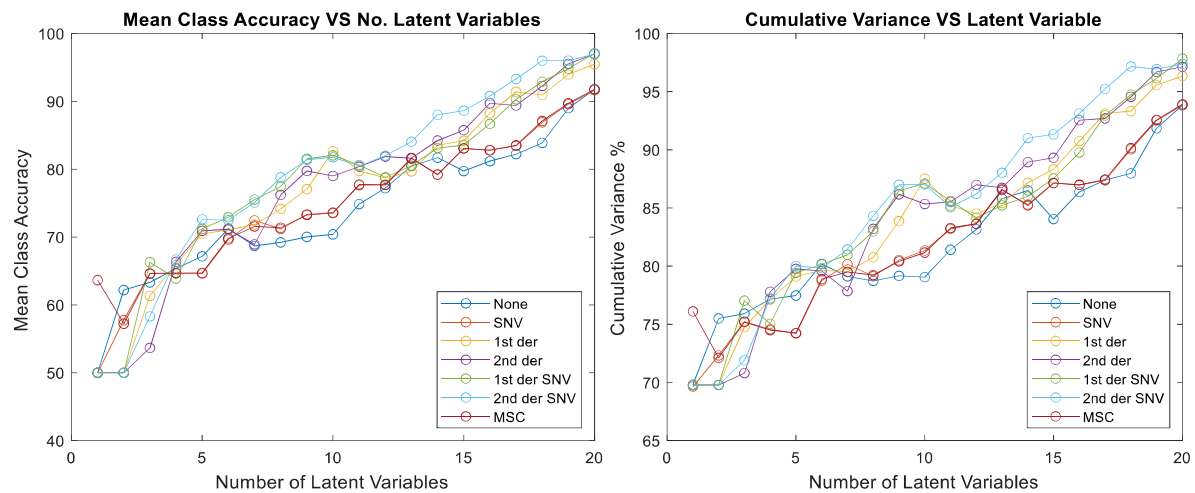


Figure 15: Germination: Mean Class Accuracy (left) and cumulative variance (right) for PLSDA with cross validation (VNIR+SWIR combined)

4.4.1.2 PLSDA Using Full Calibration and Test Set

The results for building a PLSDA model using the full calibration set and testing it using the full test set for germinating and non-germinating seeds are shown below. In fig.16 below, the correct classification percentage with regards to the number of latent variables is plotted for each pretreatment used. Surprisingly, using no pretreatment for the PLSDA model resulted in the highest correct classification percentage at 84% for only 5 latent variables. The next highest percentage was 82.3% at 3 and 4 latent variables using second derivative as the pretreatment. The percentage drops off for all pretreatments after approximately 10 latent variables.

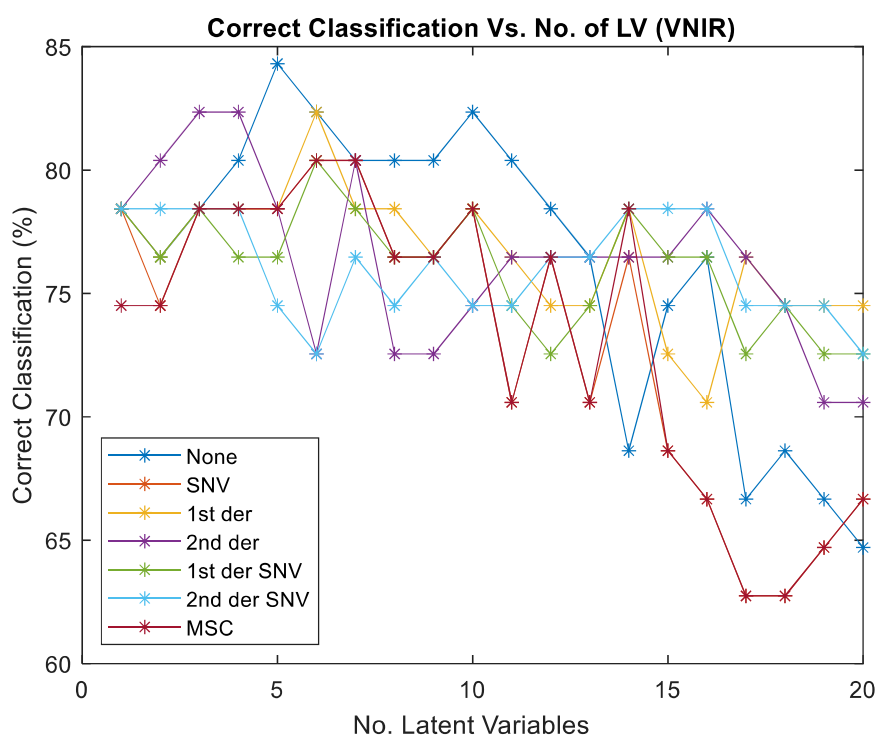


Figure 16: Germination: Correct classification from PLSDA for each latent variable with different pretreatments (VNIR)

PLSDA was then applied to the SWIR spectra. The correct classification for each pretreatment for different number of latent variables is plotted in fig.17. The optimal number of latent variables to use for PLSDA on this region would be 8, giving 82.3% correct

classification, which is the same as the VNIR region, but the VNIR region needing only 3 or 4 latent variables. Combining the two regions (fig.18) to give the full wavelength range resulted in the highest correct classification at 86.2% with no pretreatment at 6 latent variables. The next highest in the combined region was 84.3% for both first derivative with SNV at 3 latent variables, and MSC at 6 latent variables.

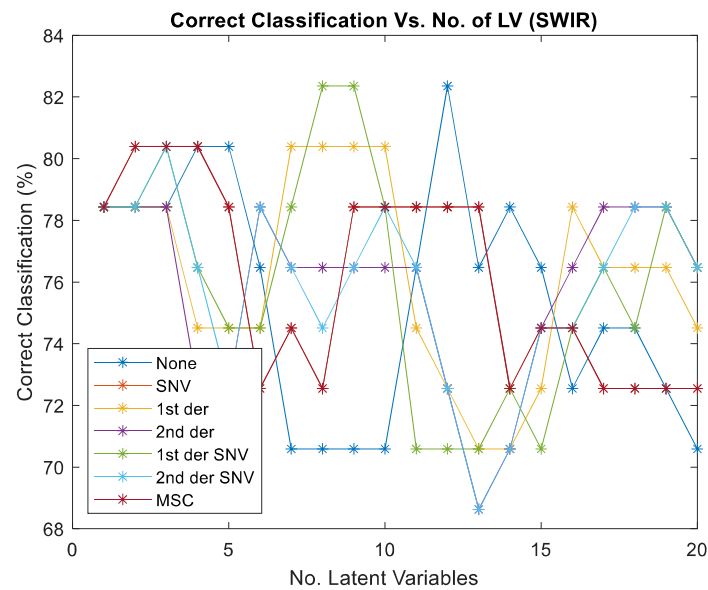


Figure 17: Germination: Correct classification from PLSDA for each latent variable with different pretreatments (SWIR)

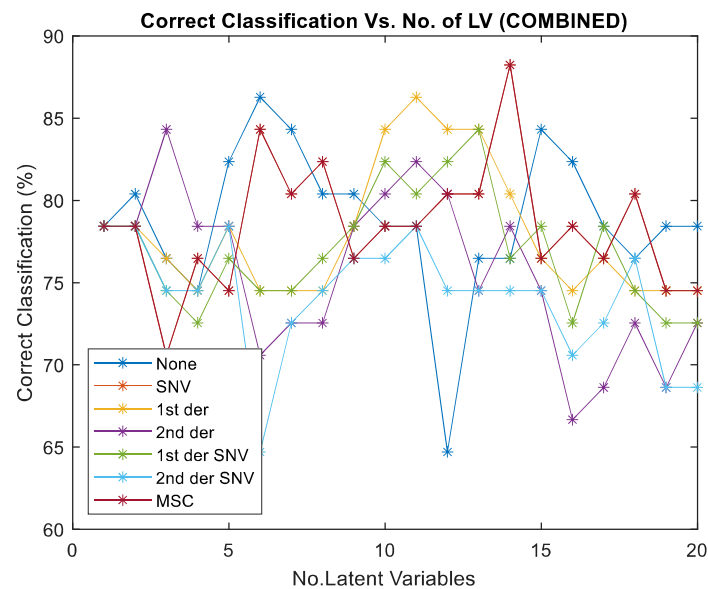


Figure 18: Germination: Correct classification from PLSDA for each latent variable with different pretreatments (VNIR+SWIR combined)

4.4.2 New VS Aged

PLSDA was also used to investigate how well hyperspectral imaging can differentiate and classify groups of old and new beans spectra. The results for this are shown below.

4.4.2.1 PLSDA with Cross Validation

PLSDA using cross validation was first used using the calibration set taken from the structure containing the beans that were not used in the germination test. The mean class accuracy for the VNIR, SWIR and combined regions as shown below in fig.19-21. The corresponding cumulative variance graphs can be seen in appendix 10. Looking at the VNIR region first, it shows that a mean class accuracy of 100% was achievable, although the number of latent variables used was quite high (14,15). The accuracy remained quite high even with the drop in latent variables, with 95% at 8 latent variables using first derivative with SNV.

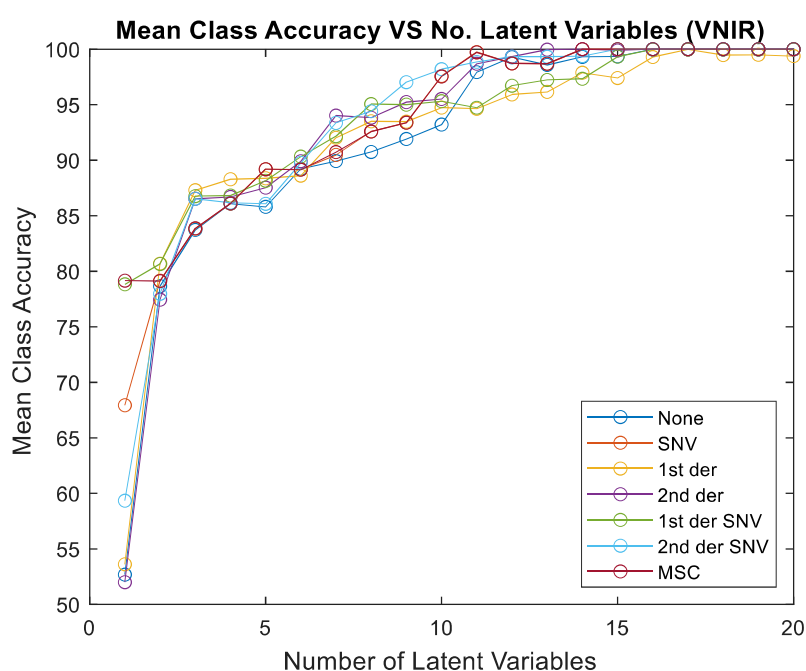


Figure 19: AGE: Mean Class Accuracy PLSDA with cross validation (VNIR)

In the SWIR region (fig.20), 100% accuracy was also achievable, but with only 9 latent variables for pretreatments of both first and second derivatives with SNV. With even less latent variables 99.9% accuracy was achievable at 7 latent variables with first derivative as pretreatment, or 99.3% accuracy with 6 latent variables using second derivative on its own. This shows that the SWIR region was more accurate than the VNIR region using PLSDA with cross validation.

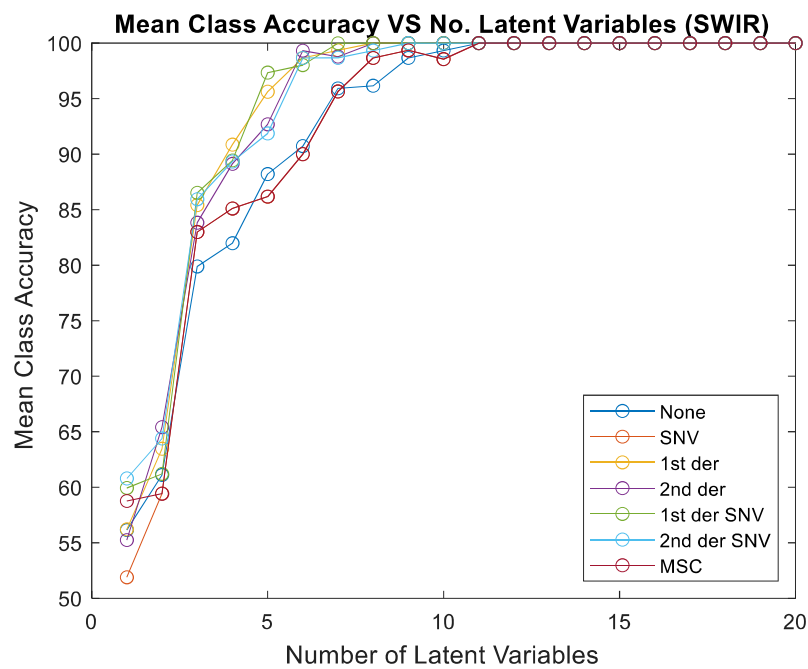


Figure 20: AGE: Mean Class Accuracy for PLSDA with cross validation (SWIR)

The results from the combined wavelength range are shown in fig.21. A high accuracy of 100% was also achievable using the full range, but the results did not improve from using only the SWIR region. This accuracy was achieved using 10 latent variables with a pretreatment of second derivative with SNV. At 6 latent variables, 92% was achieved using first derivative, which is less than that of SWIR at the same number of latent variables.

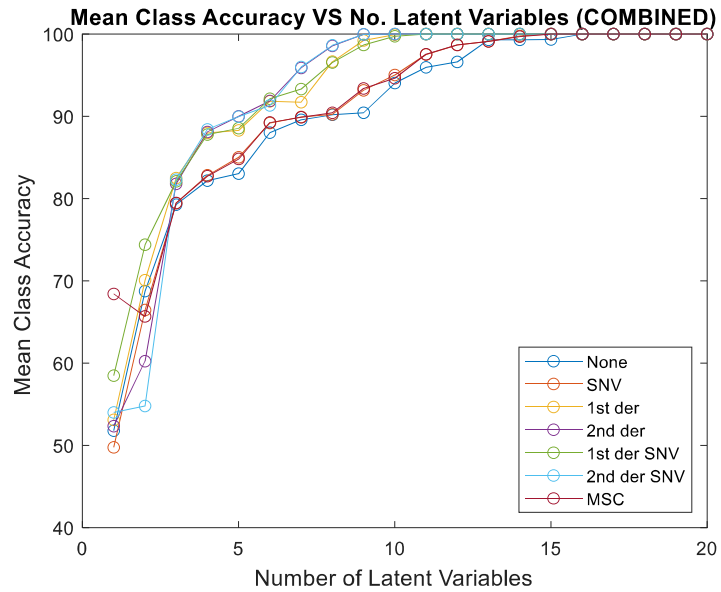


Figure 21: AGE: Mean Class Accuracy for PLSDA with cross validation (VNIR+SWIR Combined)

4.4.2.2 PLSDA with Full Calibration and Test Set

The full calibration and test sets were then used in PLSDA. The results for this are shown fig.22-24. Correct classification of 100% was achieved using the VNIR region first at 8 latent variables with first derivative, and second at 9 latent variables using MSC or SNV. Using MSC also gave correct classification of 90% for only 1 latent variable.

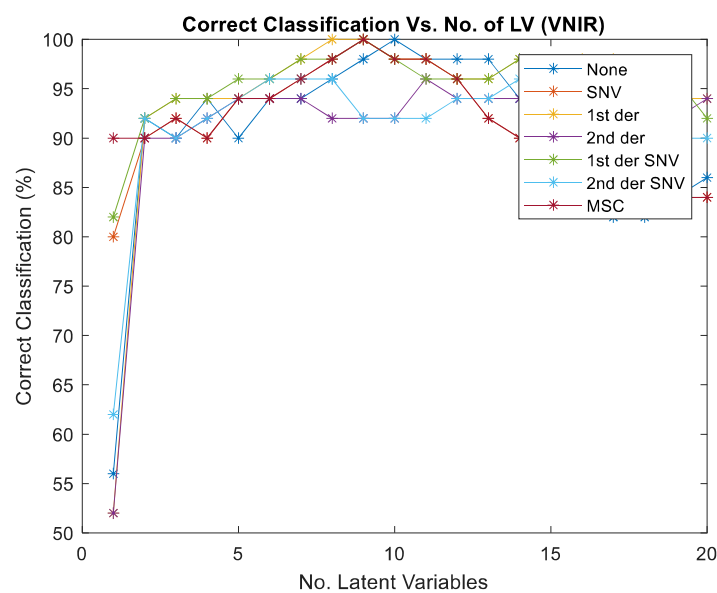


Figure 22: AGE: Correct classification from PLSDA for each latent variable with different pretreatments (VNIR)

As can be seen in fig.23, using the SWIR region for PLSDA results in much higher correct classification at a lower number of latent variables. It achieves a correct classification of 100% at only 3 latent variables for second derivative with SNV. It then achieves 100% again for no pretreatment at only 4 latent variables. MSC then remains at 100% from 6 latent variables onwards. Then looking at fig.24 for the combined wavelength range, it's clear that it performs better than PLSDA in the VNIR region, with it reaching 100% correct classification at 6 latent variables for second derivative with SNV. Throughout the PLSDA, the SWIR region seems to provide the best results for classification.

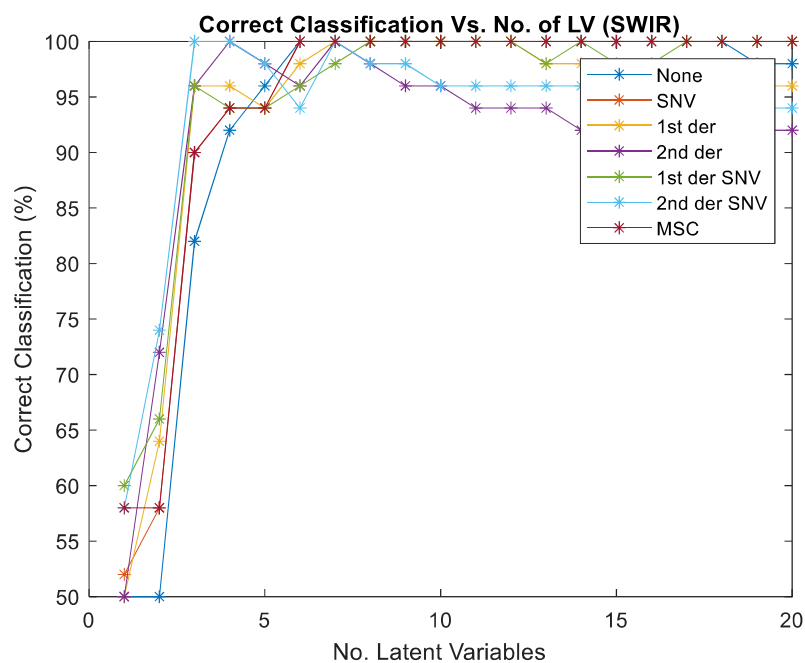


Figure 23: AGE: Correct classification from PLSDA for each latent variable with different pretreatments (SWIR)